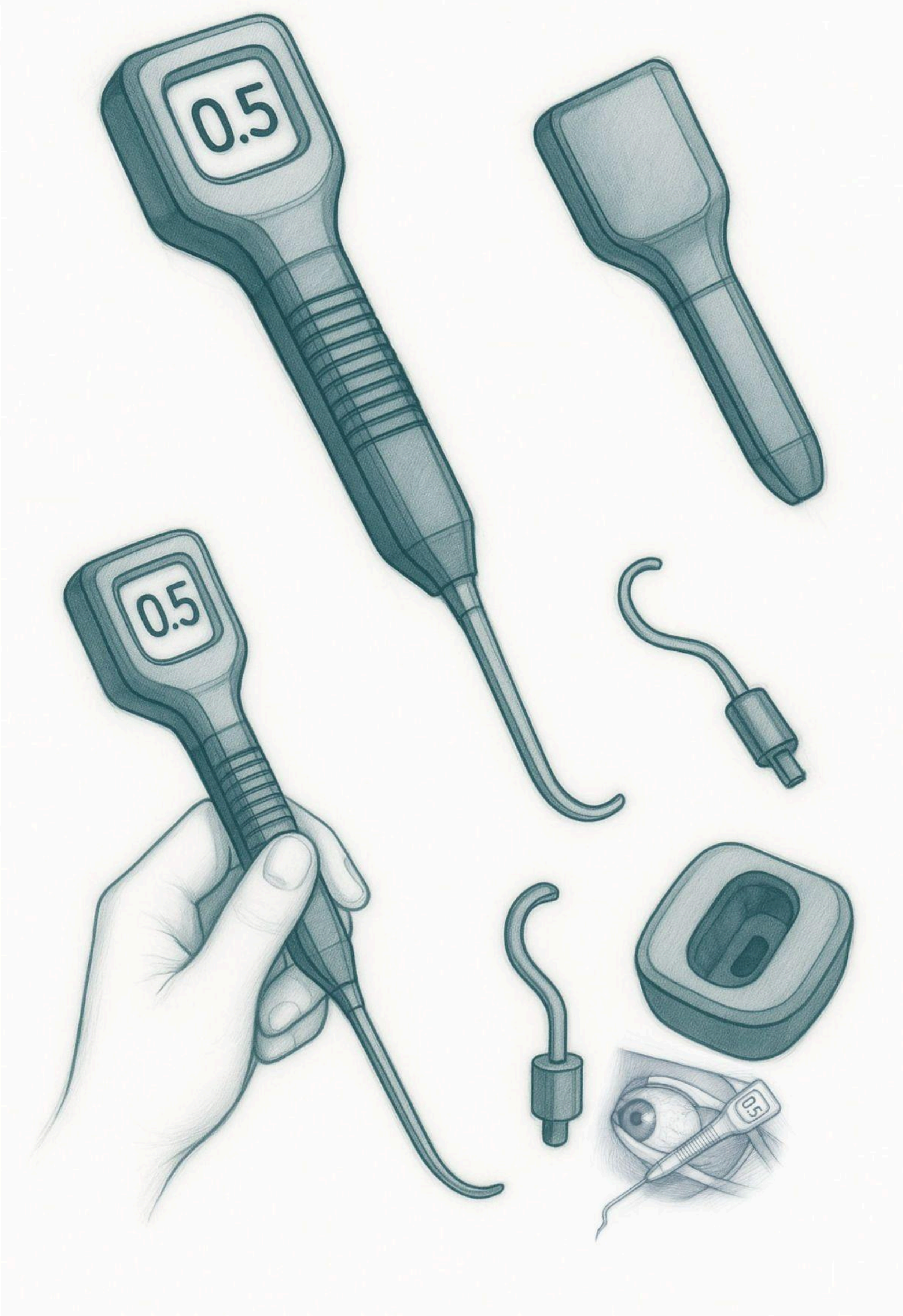

Title: Concept Sketch: Force-Sensing Rectus Muscle Hook with Digital Integrated Force Readout and Swappable Tip
Date: 2025-07-06
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Scale: Not to Scale
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Sheet: 1 of 1
Institution: UCI
Project: Dr. Suh Muscle Hook Innovation

Abstract

Strabismus surgery requires precise manipulation of extraocular muscles, particularly the rectus group, to correct ocular misalignment. A critical but currently subjective aspect of this procedure is assessing muscle tautness before and after disinsertion. Surgeons must manually pull on the extraocular muscles using a muscle hook to gauge tension or stiffness, which varies significantly depending on the surgeon's tactile experience. Unfortunately, this method offers no objective feedback, and excessive pulling may lead to muscle damage or even rupture. In cases of calcified or fibrotic muscles, this risk is heightened, resulting in a challenging and potentially dangerous surgical scenario [1].

Currently, no standardized instrument exists to quantify the force applied during muscle hooking, and there is no established threshold for what constitutes a “normal” or “pathological” pull force. While some experimental studies have attempted to measure tensile strength in extraocular muscles using load cells or spring gauges, these tools are either too bulky or not designed for sterile, real-time intraoperative use [2]. This lack of quantification not only introduces variability in surgical technique and outcome but also limits opportunities for data-driven training of new ophthalmic surgeons.

To address this unmet need, we are developing a force-sensing rectus muscle hook that integrates a miniaturized strain-gauge load cell and a digital force readout into a form factor nearly identical to the standard surgical hook. The design is ergonomic and intuitive, allowing surgeons to hold the device in a pencil grip similar to conventional muscle hooks. The integrated force gauge provides real-time feedback, displaying pull force in newtons, with a target range of 0 to 1 N. This range was determined based on preliminary values provided by Dr. Suh, where the approximate contact area of the extraocular muscle is 0.5 mm by 1.5 cm and the estimated pressure applied during pulling is 20 kPa. Using the relation $F = P \times A$, the resulting force is approximately 0.15 N. This falls well within the sub-newton scale and is appropriate for delicate ocular structures. In parallel, we are also exploring the use of a mechanical hanging scale mechanism, similar to analog spring-based devices, as an

alternative design. This approach removes the need for electronic components and offers a sterilization-friendly, low-cost solution while still providing visible, quantifiable force feedback during muscle manipulation.

To support sterilization and adaptability, the device includes a swappable hook tip using a key-and-lock mechanism at the shaft interface. This system, similar to an electric toothbrush head, allows the surgical team to replace the hook tip between procedures or customize hook geometries based on clinical needs without compromising the core force-sensing components.

Initial calibration and concept validation suggest that this device can reliably measure forces within the physiological range observed during typical strabismus surgery. By offering a quantifiable metric for muscle tautness, this tool has the potential to reduce intraoperative complications such as muscle avulsion, improve consistency across surgeons, and serve as a valuable training aid for ophthalmology residents. Standardized force data may also support surgical planning tools and simulation-based surgical training in the future.

This concept sketch represents the foundation for a future CAD-based prototype and feasibility testing phase. We believe this innovation is suitable for eventual commercialization in the ophthalmic surgical instrument market, addressing both clinical and educational needs.

References

[1] Luo, L. et al. (2017). The Suh Muscle Hook: A New Muscle Hook for Tight Extraocular Muscles during Strabismus Surgery. Open Journal of Ophthalmology, 7(1), 51–56.

[2] Shin, H.J. et al. (2025). Novel Device for Intraoperative Quantitative Measurements of Extraocular Muscle Tensile Strength. Biosensors, 15(6):347.